Recipients: Ali Kumail (Full Stack) · Ali Hafidz (UX/UI) · V3 replaces V2 — adds F8, Statistical Engine, multi-farm MVP, Report Generator

WHAT IS IT

The Protocols & Observations module is the operational heart of NuBred. It defines what is measured, how, and when — and it is the system that collects that data from Field Observers, NuBred Node sensors, and Vision AI cameras. In V3, this module also includes the Statistical Analysis Engine that transforms raw observations into genotype performance metrics and generates the data for the trial report.

> *NuBred Node IS the Field Observer. The human Field Observer is the Node operating temporarily until the physical hardware is deployed. The mobile interface, the data schema, the API endpoint, the validation workflow — all of it is Node infrastructure running with a human as the sensor. When hardware arrives, the source field changes from Manual to Node. Nothing else changes.*

THE CRITICAL DISTINCTION — UI FROZEN vs BACKEND ACTIVE

This section explains the most important architectural decision for MVP development. It must be understood by Ali Kumail before writing a single line of code.

> *FROZEN means the UI tab is not shown in the MVP interface. ACTIVE means the data model is built, the API endpoints exist, and the AI extraction pipeline populates the data. These are completely different things. A module can be UI-frozen and backend-active simultaneously — and for NuBred, most modules must be.*

Why? Because the AI extraction pipeline reads all documents simultaneously and extracts data for every module — Supply Chain, IP Rights, Quarantine, Contracts, Genotype, Phases, Protocols. If Ali Kumail builds only the entities for visible modules and skips the rest, the extraction pipeline has nowhere to put the data it finds. The result is either: (1) the pipeline crashes on unknown entities, or (2) the pipeline discards data that will need to be re-extracted later at additional cost. Both outcomes are worse than building the full data model now.

The correct mental model: the backend is always complete. The frontend shows what is ready for the user. When Ali Hafidz completes the Figma for Supply Chain, Ali Kumail activates the tab — the data is already there. No re-extraction, no migration, no technical debt.

Module / Feature	MVP Status	Backend / Data model
Field Observation flow	**☑ ACTIVE — priority 1**	Full implementation. Protocol, ProtocolParameter, ObservationSession, ObservationValue, FieldObserverAssignment. Mobile UI. GET /api/protocols/active endpoint.
Genotype section	**☑ ACTIVE**	Full implementation. Genotype entity. Linked to ObservationSession and Protocol.
Phase management	**☑ ACTIVE**	Full implementation. Phase entity, phase gate logic, Promote/Repeat/Discard flow.
AI contract extraction	**☑ ACTIVE — build now**	CRITICAL: Build the full AI extraction pipeline from day one. Claude API integration, JSON schema extraction, confidence levels. Even if the review UI is simplified for MVP, the extraction engine must be complete. See note below.

Statistical Analysis Engine	☑ **ACTIVE — build now**	Build GenotypePerformance entity and computation logic from day one. Even if the report UI is simplified, the statistical layer must run on every validation batch.
F8 AgronomicOperation	☑ **ACTIVE — data model only**	Build Prisma entity and API endpoint. UI can be a simple log form for MVP. Data must be stored from day one — it is required for the trial report.
Supply Chain diagram	◇ **UI FROZEN — backend active**	Build SupplyChainActor and SupplyChainEdge entities. AI extraction populates them. React Flow diagram UI is not shown in MVP tab bar but the data exists and is queryable.
IP Rights section	◇ **UI FROZEN — backend active**	Build IPRight entities (base + PBR/Patent/Trademark child entities). AI extraction populates them. UI tab not shown in MVP.
Quarantine section	◇ **UI FROZEN — backend active**	Build Quarantine entity. Status computed field. UI tab not shown in MVP.
Chronology section	◇ **UI FROZEN — backend active**	Chronology events generated as side effect of all document analyses and session submissions. Stored in DB. Timeline UI not shown in MVP.
Full Report Generator UI	**UI FROZEN — logic active**	Statistical engine runs. Data is ready. Full report PDF generation UI is post-MVP. VM can export raw data tables as CSV in MVP.

Note on AI extraction engine: the extraction pipeline must be complete for all modules from day one. Even if the VM only sees Genotype and Phases in the UI, the AI must extract and store Supply Chain actors, IP Right records, Quarantine requirements, and contract clause analyses every time documents are uploaded. The cost of extracting now is zero marginal cost. The cost of re-extracting later is real API cost plus development time.

MVP FLOW — KIWI FIRST CLIENT

The first real client is a kiwi project with 12 genotypes across multiple farms. This is the flow that NuBred must execute correctly for the MVP. Everything else is secondary to making this flow work end-to-end.

#	Actor	Action	NuBred result
1	Variety Manager	Uploads kiwi trial contract + observation protocol document to NuBred.	AI extracts: 12 genotypes, Trial phase, protocol parameters, Field Observer obligations. Project created in Draft.
2	Variety Manager	Reviews AI extraction section by section. Confirms genotypes, phases, and protocol. Assigns Field Observers to each farm.	Project activated. Field Observer assignments created per genotype per farm.
3	Field Observer	Opens NuBred mobile app. Sees list of assigned observation sessions for today. Selects Farm 1, selects genotype, opens observation form.	Form generated dynamically from active protocol — correct parameters, input types, and frequencies for this phase.
4	Field Observer	Fills observation form: vigour scale, phenology BBCH stage, DM% sample, firmness, notes. Captures photos. Submits.	ObservationSession created with source=Manual. All values stored. Session status: Submitted.
5	Field Observer	Repeats for all 12 genotypes in Farm 1. Then moves to Farm 2 and Farm 3. Offline if no connectivity — syncs when back online.	Sessions accumulate per genotype per farm per date. Full dataset builds over the season.
6	Variety Manager	Reviews submitted observations on web interface. Sees all 12 genotypes across 3 farms. Validates sessions, flags anomalies, requests re-observation if needed.	Sessions marked Validated or Rejected. Validated data feeds the Statistical Analysis Engine.
7	NuBred System	Statistical Analysis Engine aggregates validated observations. Calculates per-genotype means, rankings, and phase gate scores.	Genotype performance dashboard updated. Phase gate eligibility computed. Report data ready.
8	Variety Manager	Reviews genotype performance. Makes Promote/Repeat/Discard decisions. Generates trial report.	Phase gate executed. Report generated automatically from aggregated data.

The VM's most important view in the kiwi MVP is a genotype × farm matrix — 12 rows (genotypes), N columns (farms), cells showing the aggregated performance score for each combination. This is what drives the Promote/Repeat/Discard decision. Ali Hafidz: this matrix is the centerpiece of the Phases section UI.

FIELD OBSERVER ROLE — V3 UPDATES

Field Observer is a functional role added to any existing NuBred account. A user can be Grower AND Field Observer in the same project. V3 adds multi-farm assignment — a Field Observer can be assigned to different genotypes on different farms within the same project.

Aspect	V3 definition
What they see	Their assigned observation sessions for today. Organized by farm → genotype. One session = one genotype on one farm on one date. Not the full project, not other observers' sessions.
What they can do	Submit observation sessions for assigned genotypes/farms. Upload photos. View their own session history. Log F8 agronomic operations for their assigned plots.

Multi-farm assignment	A single Field Observer can be assigned to Farm 1 genotypes 1-6 and Farm 2 genotypes 7-12 simultaneously. FieldObserverAssignment has farmId and genotypeId — the combination defines the scope.
Offline mode	Mandatory from day one. App stores active protocol and pending sessions in IndexedDB. Field Observer completes sessions offline. Data syncs automatically when connectivity is restored. Conflict detection if same session submitted twice.
NuBred Node relationship	The Field Observer mobile interface is the Node control interface. When Node hardware is installed, this same app is used to review Node-transmitted sessions, validate anomalies, and intervene manually when sensors cannot measure something.

OBSERVATION FORM — DYNAMIC GENERATION

The observation form shown to the Field Observer is never a static form. It is generated dynamically from the active protocol for the current phase and genotype. This means the same app shows completely different forms for a kiwi DM% measurement in Trial phase vs a strawberry Brix measurement in Pilot phase.

1 **Field Observer opens app and selects session**
App calls GET /api/protocols/active?projectId=X&genotypeId=Y&farmId=Z&phaseId=W. Returns the active protocol JSON with all parameters for this combination.

2 **App generates form from protocol JSON**
For each ProtocolParameter in the response: render the correct input component (Number field, Scale selector, Option dropdown, Distribution histogram, Date picker, Photo capture, Text area). Apply validation rules from scaleMin/scaleMax, threshold.

3 **Field Observer completes and submits**
Values validated client-side. Eliminatory threshold breaches flagged immediately with a warning — Field Observer must acknowledge before submitting. Session submitted as ObservationSession with status=Submitted.

4 **Backend validates and flags eliminatory breaches**
On session receipt: compute eliminatoryFlagged for each value. If any L1 or L2 breach detected: flag GenotypePerformance immediately, notify VM, set phaseGateEligible=false.

5 **VM validates session**
VM sees submitted sessions in web interface. Reviews values, photos, notes. Validates or rejects. On validation: trigger Statistical Analysis Engine for this genotype + farm combination.

STATISTICAL ANALYSIS ENGINE

The Statistical Analysis Engine is the layer that transforms raw ObservationValues into the GenotypePerformance entity. It runs automatically after every VM validation batch. It produces the data that feeds both the VM's decision dashboard and the trial report generator.

Computation	Definition

Per-parameter mean per genotype	Arithmetic mean of all validated scalar ObservationValues for that parameter + genotype combination across all sessions in the phase. For Distribution values (calibre), weighted mean per size class.
Standard deviation	Standard deviation across sessions for each parameter per genotype. Indicates consistency of performance — high SD means variable performance across dates or farms.
Per-farm breakdown	Same statistics computed per farm. Enables VM to see if a genotype performs consistently across locations or shows site-specific behavior. Critical for multi-farm MVP like kiwi.
Ranking within phase	Genotypes ranked by weighted score within the same phase at the same site. Ranking is recalculated after each VM validation batch — it is always current.
Weighted score	Composite score calculated from weighted combination of non-eliminatory parameters using the species scoring profile. Eliminatory parameters are checked separately — a breach disqualifies regardless of weighted score.
Eliminatory flag propagation	If any ObservationValue for a genotype breaches an eliminatory threshold (L1, L2, or L3), the GenotypePerformance record is immediately flagged. VM sees a red alert on that genotype. phaseGateEligible = false.
Benchmark comparison	If a benchmark variety is defined in the project (from the Benchmark Registry), all quantitative relative parameters are computed as % vs benchmark mean. Delta EBITDA calculation uses this comparison.

The Statistical Analysis Engine is NOT a UI feature — it is a backend computation job. It runs silently after validation and updates the GenotypePerformance table. The VM sees the results through the genotype performance dashboard and the phase gate indicators. Ali Kumail: implement as a NestJS background job triggered by the session validation event.

F8 AGRONOMIC OPERATIONS — IMPLEMENTATION

F8 is the operational log of all agronomic interventions during the trial. It is mandatory for generating a complete trial report and for calculating the cost side of delta EBITDA. In MVP it is a simple log form accessible from the Field Observer app and the VM web interface.

UI requirement	Spec
Log entry form	Date, operation type selector (Irrigation / Fertilization / Pest Treatment / Pruning / Soil Management / Other), product name, active substance, dose + unit, volume (for irrigation), plot/farm selector, notes. Simple form — not complex.
Log viewer	Chronological list of all F8 entries for the project, filterable by farm, plot, operation type, and date range. VM and Field Observer both have access.
Report integration	Section 3 of the trial report (Agronomic Information) is populated directly from F8 log. Treatment table format identical to ALSIA report: Date | Product | Active substance | Dose. Irrigation section: volumes and intervals per period.

TRIAL REPORT GENERATOR — WHAT NUBRED AUTO-GENERATES

The trial report — like the ALSIA Final Report for SanLucar strawberry — is the primary output of the Observational Protocol. NuBred generates it automatically from the validated data. The following table shows exactly what is generated and what it requires.

Report section	NuBred source	Auto-generated?	Requires

1. Study objective	Phase record — objective field	☑ **Yes — Claude narrative**	Phase confirmed by VM
2. Location & general info	Project record, Supply Chain, Farm entity	☑ **Yes — structured fields**	Farm data in project
3. Agronomic information	F8 AgronomicOperation log	☑ **Yes — from F8 log**	F8 data entered during trial
4. Trial design & protocol	Protocol + ProtocolParameter + Genotype	☑ **Yes — structured fields**	Protocol confirmed by VM
5. Phenological observations	ObservationValues — F1 parameters	☑ **Yes — tables + charts**	F1 sessions validated
6. Results — production	GenotypePerformance — F3 family	☑ **Yes — tables + cumulative charts**	F3 sessions validated + stats engine
6. Results — fruit quality	GenotypePerformance — F2 family	☑ **Yes — tables + rankings**	F2 sessions validated + stats engine
6. Results — shelf life	ObservationValues — F5 family, multi-interval	☑ **Yes — tables per interval**	F5 sessions validated
6. Results — organoleptic	ObservationValues — F2 organoleptic parameters	☑ **Yes — panel summary table**	F2 panel sessions validated
6. Statistical discussion	GenotypePerformance — all families	☑ **Yes — Claude narrative from stats**	Stats engine computed + VM review
7. Data conservation	System boilerplate + project metadata	☑ **Yes — auto-generated**	None
8. Study validation	VM signature + validation timestamp	☑ **Yes — VM approves report**	VM explicit report approval action
Annex A — Climate data	ObservationValues — F6 family, daily	☑ **Yes — full daily table**	F6 data logged daily
Annex B — Soil analysis	AgronomicOperation — soil type (uploaded document)	⚠ **Partial — if document uploaded**	Soil analysis document uploaded to project
Annex C — Photos	ObservationValue.photoUrl — all sessions	☑ **Yes — auto-selected by VM**	Photos captured during field sessions
Annex D — Accreditation	Project record — research center + certification reference	☑ **Yes — from project metadata**	Accreditation details in project

The statistical discussion narrative (Section 6 conclusions) is generated by Claude from the GenotypePerformance data. Claude receives the ranked means, standard deviations, eliminatory flags, and benchmark comparisons — and generates a 300-500 word professional narrative in the style of an agronomic research report. VM reviews and approves before the report is finalized.

BACKEND DATA MODEL — V3

Entity	Key fields	Notes
Protocol	id, projectId, phaseId, protocolType (Observational | Commercial), name, objective, category (F1-F8), speciesId, citedSource	protocolType is new in V3 — distinguishes Observational from Commercial. One

		protocol per phase per parameter family. A project may have multiple protocols per phase.
ProtocolParameter	id, protocolId, name, family (F1-F8), inputType, scaleMin, scaleMax, scalePolarity, howToMeasure, unit, frequencyType, frequencyValue, threshold, thresholdType, eliminatoryLevel (L1|L2|L3|null), isCustom, sampleSize	scaleMin and scaleMax replace hardcoded 1–5. scalePolarity: higher_better | lower_better. eliminatoryLevel null means non-eliminatory. isCustom flags VM-added Level 3 parameters.
ObservationSession	id, protocolId, genotypeId, plotId, farmId, observerId, source (Manual|Node|Vision), nodeId, cameraId, modelVersion, sessionDate, status (Draft|Submitted|Validated|Rejected), submittedAt, validatedAt, validatedBy	source field is the only difference between Manual, Node, and Vision sessions. farmId and plotId added in V3 for multi-farm kiwi MVP. modelVersion tracks Vision AI model version for audit.
ObservationValue	id, sessionId, parameterId, scalarValue, structuredValue (JSON), photoUrl, note, recordedAt, eliminatoryFlagged, eliminatoryLevel	structuredValue (JSON) is new in V3 — stores Distribution input type values (calibre histogram). eliminatoryFlagged computed on insert: if value breaches eliminatory threshold, flag immediately and notify VM.
FieldObserverAssignment	id, projectId, userId, genotypeId, farmId, plotId, protocolId, assignedBy, assignedAt, activeFrom, activeTo	farmId and plotId added in V3 for multi-farm MVP. A Field Observer can be assigned to different genotypes on different farms. activeTo allows temporary assignments.
AgronomicOperation	id, projectId, phaseId, farmId, plotId, operationDate, operationType (Irrigation|Fertilization|PestTreatment|Pruning|SoilManagement|Other), product, activeSubstance, dose, doseUnit, volume, notes, loggedBy	New entity in V3. Covers F8 Agronomic Operations. operationType determines which fields are relevant. product and activeSubstance for chemical

		treatments. volume for irrigation. Linked to farm and plot to enable per-plot agronomic context in the trial report.
GenotypePerformance	id, projectId, phaseId, genotypeId, parameterFamilyId, calculatedAt, meanValue, stdDev, n (sample count), minValue, maxValue, rankInPhase, weightedScore, eliminatoryFlagged, phaseGateEligible	New entity in V3. Output of the Statistical Analysis Engine. Computed from validated ObservationValues. Updated after each VM validation batch. phaseGateEligible computed from eliminatoryFlagged + weightedScore vs threshold.

API ENDPOINTS — MVP PRIORITY

GET /api/protocols/active

Returns the active protocol JSON for a given project + genotype + farm + phase combination. This is the endpoint that generates the Field Observer observation form AND the endpoint that NuBred Node hardware will call when deployed. Must be built in MVP. Parameters: projectId, genotypeId, farmId, phaseId.

POST /api/observations/session

Receives a completed ObservationSession from Field Observer (mobile) or NuBred Node (hardware) or Vision system (AI camera). Validates all values, computes eliminatory flags, stores session with Submitted status, notifies VM. Body: full session JSON including all ObservationValues.

PATCH /api/observations/session/:id/validate

VM validates or rejects a submitted session. On validation: triggers Statistical Analysis Engine for affected genotype + farm. On rejection: session status → Rejected, notification sent to Field Observer.

GET /api/genotypes/:id/performance

Returns GenotypePerformance data for a genotype — means, rankings, eliminatory flags, phase gate eligibility. Used by the VM dashboard and the phase gate UI.

POST /api/observations/agronomic

Logs an F8 AgronomicOperation. Available to both Field Observer and VM. Body: operationDate, operationType, product, dose, farmId, plotId, notes.

POST /api/reports/trial

Generates the trial report for a phase. Triggers: (1) data aggregation from all validated sessions, (2) Claude narrative generation for sections 1, 6-discussion, and 7-8, (3) PDF assembly. Response: report draft for VM review and approval.

DECIDED — DO NOT REOPEN

- FROZEN means UI not shown. ACTIVE means backend built. All modules are backend-active from day one.
- AI extraction pipeline is complete from day one — all entities must exist in Prisma before first document upload
- Statistical Analysis Engine runs after every VM validation batch — GenotypePerformance updated automatically
- F8 AgronomicOperation is mandatory — required for trial report Section 3 and delta EBITDA cost side
- Field Observer has farmId in assignment — multi-farm support is in MVP (kiwi requirement)
- GET /api/protocols/active is the shared endpoint for Field Observer mobile, NuBred Node, and Vision AI
- Trial report is auto-generated from validated data — Claude generates narrative sections
- Three data sources: Manual / Node / Vision — same schema, same endpoints, source field only difference
- ObservationValue supports both scalarValue and structuredValue (JSON) for Distribution input type

OPEN — TEAM TO IMPLEMENT

- All Prisma entities from data model table above — Protocol, ProtocolParameter, ObservationSession, ObservationValue, FieldObserverAssignment, AgronomicOperation, GenotypePerformance
- GET /api/protocols/active endpoint — priority 1 for MVP
- Mobile observation form — dynamic generation from protocol JSON, all input types, offline-capable
- Statistical Analysis Engine — background NestJS job triggered by session validation
- GenotypePerformance computation — mean, std dev, weighted score, rank, eliminatory flag, phase gate eligibility
- F8 log form — simple entry form for VM and Field Observer
- Trial report generator — data aggregation + Claude narrative + PDF assembly
- Multi-farm FieldObserverAssignment — farmId in assignment entity and in session

DO NOT

- Do not build only the entities for visible MVP modules — all entities must exist from day one
- Do not call the AI extraction pipeline twice when modules are 'unfrozen' — it must extract everything on first upload
- Do not build a different schema for Node or Vision sessions — source field only
- Do not make offline mode optional for the Field Observer mobile app
- Do not skip GenotypePerformance entity — it is the output the VM needs to make decisions
- Do not generate the trial report narrative before VM validates all sessions — unvalidated data must not appear in reports